

- 2 (a) (i) (air) resistance increases with speedM1
 resultant / accelerating force decreases A1 [2]
- (ii) *either* (air) resistance is zero
 or weight / gravitational force is only force B1 [1]
- (b) use of gradient of a tangentM1
 acceleration = $1.9 \pm 0.2 \text{ m s}^{-2}$ A2 [3]
(for values $> \pm 0.2$ but ≤ 0.4 , allow 1 mark)
(answer 3.3 m s^{-2} scores no marks)
- (c) (i) 1 weight = $90 \times 9.8 = 880 \text{ N}$ A1 [1]
(use of $g = 10 \text{ m s}^{-2}$ then deduct mark but once only in the Paper)
 2 accelerating force = $90 \times 1.9 = 170 \text{ N}$...(allow ecf) A1 [1]
- (ii) resistive force = $880 - 170 = 710 \text{ N}$ A1 [1]
(allow ecf but only if resistive force remains positive)

[Total: 9]

- 3 (a) (i) *either* sum / total momentum (of system of bodies) is constant